

Ziyu (Simon) Xiao

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EDUCATION

Georgia Institute of Technology	Atlanta, GA
M.S. in Electrical & Computer Engineering, GPA: 3.65/4.0	08/2023 - 05/2026
University of Illinois Urbana-Champaign	Champaign, IL
B.S. in Electrical Engineering, GPA: 3.63/4.0	08/2019 - 05/2023
Zhejiang University <double degree>	Hangzhou, China
B.E. in Electrical Engineering and Automation, GPA: 3.90/4.0	09/2019 - 06/2023

SKILLS

ML & AI: PyTorch, LLMs, RAG, Graph-RAG, Prompt Engineering, Transformers, Diffusion Models, RL, Imitation Learning, Representation Learning, Multimodal Learning, VLMs, 3D Vision, Search Recommendation System

Robotics: Isaac Gym/Sim, NVIDIA cuRobo, ROS, OpenCV, Motion Planning

Software: Python, SQL, Spark, Hive, MySQL, Neo4j, Cypher, CUDA, Linux, Shell, Docker, Git, C/C++, Java, Spring Boot, Redis, Kafka

PUBLICATION

Mnemis: Dual-Route Retrieval on Hierarchical Graphs for Long-Term LLM Memory

Zihao Tang, ****Ziyu Xiao****, Zengxuan Wen, Zelin Li, Jiayi Zhou, Hualei Wang, Haohua Wang, et al., Xin Yu*, Microsoft*
Accepted by ACL 2026. (top-tier NLP venue) <<https://arxiv.org/abs/2602.15313>>

Going with the Flow: Koopman Behavioral Dynamics as Implicit Visual-Motor Planners for Dexterous Manipulation

Yunhai Han, ****Conference on Robot Learning (CoRL) 2026. (top-tier robotics conference)** <<https://arxiv.org/abs/2602.07413>>

Video2Sim2Real: Learning Generalizable Dexterity from a Single Human Manipulation Video

Yunhai Han, Jianuo Qiu, Linhao Bai, ****Ziyu Xiao****, et al., Harish Ravichandar*

Under review of Conference on Robot Learning (CoRL) 2026. (top-tier robotics conference) <<https://arxiv.org/abs/2606.08828>>

Bi-DexHands: Bimanual Dexterous Manipulation via Reinforcement Learning

Zhaodong Yang, ****Ziyu Xiao****, et al., Harish Ravichandar*

Under review of IEEE Robotics and Automation Society (RAL) 2026. (top-tier robotics journal).

WORKING EXPERIENCE

Tiktok - Machine Learning Engineer (Recommendation-Search Group)

05/2026 – Present

Tech Stack: Python, SQL, Spark, Hive, ANN Retrieval, Swing, Two-Tower Retrieval, Multi-Head Self-Attention, Recommendation Systems

Search Recommendation / Cross-Domain User Interest Modeling

- Worked on** cross-domain interest generalization for post-search recommendation, leveraging video-domain consumption signals to retrieve videos that could stimulate users' search interests.
- Implemented** a Swing-based Video U2U2I recall pipeline to identify similar users from video co-consumption behavior and expand search-interest-related candidates from their high-value video interactions.
- Contributed** to a two-tower U2I retrieval model, using Multi-Head Self-Attention to encode recent user behavior and ANN-indexed offline Item Embeddings for efficient retrieval.
- Conducted** offline recall evaluation and pipeline diagnosis using recall coverage, coarse/fine ranking pass-through rate, search posterior metrics, and user/item coverage to identify bottlenecks in cross-domain recommendation.

Microsoft - Machine Learning Engineer Intern (Bing Ads Group)

05/2025 – 08/2025

Tech stack: Python; Neo4j, Cypher, MYSQL; LLM, Azure OpenAI (GPT); RAG; Git

Segmentation hierarchical Knowledge Graph-Enhanced Memory System for Bing Search

- Built** a long-term LLM memory system that organizes conversational history into base graphs and hierarchical graphs for similarity-based retrieval and top-down global memory selection.
- Designed** temporally-aware Neo4j / Cypher graph schemas and LLM-based construction pipelines for episodes, entities, edges, summaries, timestamps, embeddings, and deduplication.
- Developed** a dual-route Graph-RAG pipeline combining System-1 similarity search with System-2 global selection over hierarchical semantic categories.
- Achieved** SOTA performance on long-term memory benchmarks, reaching 93.9 on LoCoMo and 91.6 on LongMemEval-S; paper accepted by ACL 2026.

Xiaomi - Autonomous Driving Machine Learning Engineer Intern

01/2025 – 05/2025

Tech stack: Python; PyTorch; CUDA; OpenCV; NumPy; Diffusion models, ViT, Linux, Git

Free-Trajectory Street Novel View Synthesis With LiDAR Point Clouds and Diffusion Models

- Reproduced** FreeVS and developed a vision-only street-view novel view synthesis pipeline, replacing LiDAR/GT point clouds with MAST3R-synthesized 3D point clouds.
- Improved** temporal consistency of diffusion-generated driving videos by integrating adjacent and long-range frame context into the model architecture.

RESEARCH

Research Assistant - Georgia Institute of Technology

08/2025 – Present

STAR Lab - Advised by Prof. Harish Ravichandar

Tech stack: Python, PyTorch, Isaac Gym, Isaac Sim, NVIDIA cuRobo, CUDA, RL, Imitation Learning, Motion Planning

- Conducted research on video-to-robot learning for task-agnostic dexterous manipulation, integrating video representation learning, motion planning, and downstream RL policy training.
- Replaced object-flow features with pretrained DynaMo motion embeddings and evaluated across simulation and real-world benchmarks to analyze the effectiveness of learned video dynamics for manipulation policy learning.
- Developed a cuRobo-based collision-aware trajectory generation pipeline for dexterous hands, converting human manipulation videos and generated tabletop scenes into robot-executable trajectories for Video2Sim2Real learning.